Answers and talk notes

Determine the Area of Parallelograms and Rhombuses

What your student is learning

Tonight we find the area of a rhombus — and see why the parallelogram formula we learned in Part 1 still works on it.

Answers

Ask your student to explain before you look.

1. **B.** It is a parallelogram with four equal sides. It is a parallelogram with four equal sides: that is what a rhombus is.
2. **D.** 35 cm^2. A = b × h = 7 × 5 = 35 cm^2.
3. **A.** 80 in^2. Use a side as the base: A = 10 × 8 = 80 in^2.
4. **C.** 9, the straight-up distance. The height is 9, the straight-up distance that meets the base at a right angle.
5. **B.** 108 in^2. A = 12 × 9 = 108 in^2.

Talk about it

A rhombus has four equal sides. Which measurement is its height, and why is the side not the height?

Home mission

A different situation: lay four pencils of the same length into a square. Gently push one corner so it leans into a diamond. The sides stay the same. Which measure got smaller: the side, or the straight-up height?

When they are stuck, ask

- Is that measurement straight up from the base, or slanted?
- A rhombus is which bigger family of shapes?
- What unit does an area answer use?

Try not to say

- Multiply the two side lengths, that is the area.
- Any number on the picture can be the height.
- The units do not matter for area.